

4 (**short**) Two columns, each of length 2 m, are compressed in a testing machine as shown in Fig. 4. The ends of the columns are firmly embedded in stiff grips, and the descent of the upper grip is controlled so that the two columns always have the same length. The cross-sections of the two columns are shown in the figure. Both are made from steel with Young's Modulus of 210 GPa.

(a) Calculate the second moments of area for the two columns, using the minor axis where there is a choice. [4]

(b) How far will the upper grip have descended before one of the columns buckles? [6]

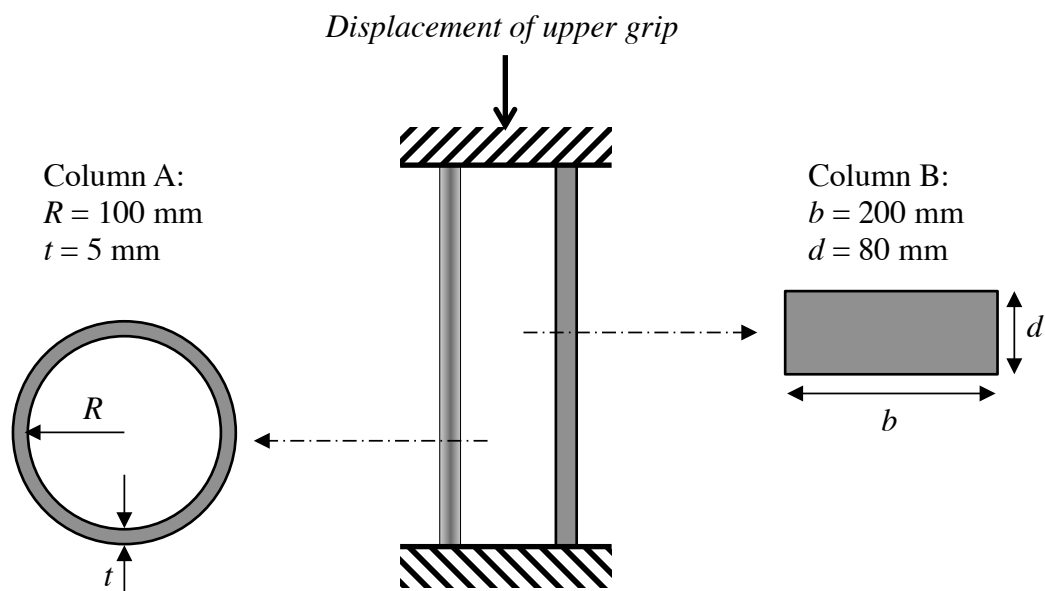


Fig. 4

NOT TO SCALE